

LING 411 Problem Session: Hypothesis Formation

Reading: Culbertson, J. & Adger, D. (2014). Language learners privilege structured meaning over surface frequency. *PNAS* 111(16), 5842–5847.

1. The parts of an argument

Beavers & Sells say that a complete argument in linguistics has four parts:

- (a) **A clear hypothesis H.** State H so that it is clear what it predicts: which data should occur and which should not. An *if... then* statement often helps.
- (b) **The data H is meant to explain.** Describe exactly which facts or patterns H covers, and leave out details that are not relevant.
- (c) **How H accounts for the data.** Show step by step why the patterns follow from H.
- (d) **A comparison with an alternative H'.** Show that H is better than a clearly stated alternative. H' can be a competing analysis or the null hypothesis (for example, “there is no such structure”).

Showing that H is *compatible* with the data is not enough. H and H' must make *different predictions*, and the data must show which prediction is correct.

2. Let's read the paper

You do *not* need to read the whole paper. You can focus on the highlighted parts, which are the parts that state the hypotheses, the predictions, and the results.

3. Key terms

- **Scope.** In *these two red cars*, the adjective combines with the noun first, then the numeral, then the demonstrative: [Dem [Num [Adj N]]].
- **Scope-isomorphic order.** Adj is closest to N, then Num, then Dem. English Dem-Num-Adj-N is scope-isomorphic, and so is its mirror image N-Adj-Num-Dem.
- **Transitional probability.** How likely one word is to follow another. In English, *this* is often followed by an adjective, but an adjective is rarely followed by *this*.

4. Questions

Q1. What is the research question of the paper? The paper names two hypotheses (Hypothesis 1 and Hypothesis 2). State each in your own words, and say which one is H (the one the authors argue for) and which one is H'.

Q2. Fill in the table: which order does each hypothesis predict at test?

Training → Test	H' (surface)	H (scope)
N-Adj, N-Dem → {N, Adj, Dem}		
N-Num, N-Dem → {N, Num, Dem}		
N-Adj, N-Num → {N, Adj, Num}		

Q3. Why did each training phrase contain only *one* modifier? What would go wrong if learners had heard phrases like *vase purple those*?

Q4. Why did the authors put the modifiers *after* the noun? Would the experiment work if they came *before* the noun, as in English?

Q5. Which hypothesis do the results support? Is H' fully ruled out?

Q6. Think further. Propose another alternative, H'', that could also explain the results without scope. How could we test it?